Abhishek Bhattacharyya[a], Daniel Rittel[b], Guruswami Ravichandran[a]
[a]Department of Aeronautics and Mechanical Engineering, California Institute of Technology, Pasadena, USA
[b]Department of Mechanical Engineering, Technion-IIT, Haifa, Israel

Strain rate dependency on deformation texture for pure polycrystalline tantalum

The deformation texture of pure polycrystalline tantalum was characterized in the range of strain rates from $10^{-3} \sim 10^{+3}\ s^{-1}$ at room temperature. Quasi-static tests were carried out on an MTS servohydraulic machine, whereas dynamic tests were carried out using a split Hopkinson pressure bar. Texture measurements were done using the electron backscattered diffraction set-up in the scanning electron microscope chamber. It was observed that the deformation texture was unaffected qualitatively by the strain rate variations, and was correlated to the insensitivity of the hardening behavior at different strain rates. Strain softening at high strain rate seemed to sharpen the texture.

Keywords: Tantalum; Texture; Shear compression specimen (SCS); Strain rate; Strain hardening

1. Introduction

Crystallographic texture, defined as the distribution of orientation of crystals (or grains) in a sample, plays a significant part in the material response and in influencing the anisotropy of macroscopic physical quantities. Texture studies improve our understanding of the underlying crystal plasticity behavior and thereby help in developing crystal-based plasticity models. So far, researchers have carried out texture analyses as a function of composition [1, 2], strain [2, 3], temperature [4, 5] and moderate strain rates from $10^{-3}\ s^{-1}$ to $10^{-1}\ s^{-1}$ [6–8]. However, there have been only a few independent studies done at very high strain rates of the order of $10^{+3}\ s^{-1}$ [9, 10]. It is very important to characterize textural response over large strain rates as such endeavors will lead to validating and widening the range of applicability of the plasticity models which are mostly strain-rate dependent [11, 12]. Consequently, this work addresses texture measurements (at room temperature) as a function of the strain rate. In the present case, pure polycrystalline tantalum (Ta) was selected as a typical bcc material whose mechanical properties are well documented. Texture analysis was done following quasi-static and dynamic deformation tests to a specific strain.

In order to achieve large strains and strain rates using the same sample geometry (and without the need of re-machining), the recently developed shear compression specimen (SCS) was used throughout the experiment (Fig. 1) [13–15].

Such a specimen is fabricated by machining two identical slots on either side of the specimen at 45° to the loading axis, for achieving maximum shear in the gage region. Under normal compression, the gage section experiences a shear dominant deformation while the bulk of the specimen remains undeformed. Large strains (with engineering strains exceeding 1.0) can be accomplished with the proper choice of gage dimensions. Similarly, strain rates ranging from $10^{-4}\ s^{-1}$ (quasi-static) to $10^{+4}\ s^{-1}$ (dynamic) can also be easily reached using this specimen. The SCS has been used for studies on oxygen-free high conductivity copper (OFHC Cu) [15], 1018 cold-rolled steel [16], and a tungsten-base heavy alloy [17]. A detailed numerical study of the SCS can be found in Dorogoy and Rittel [18, 19].

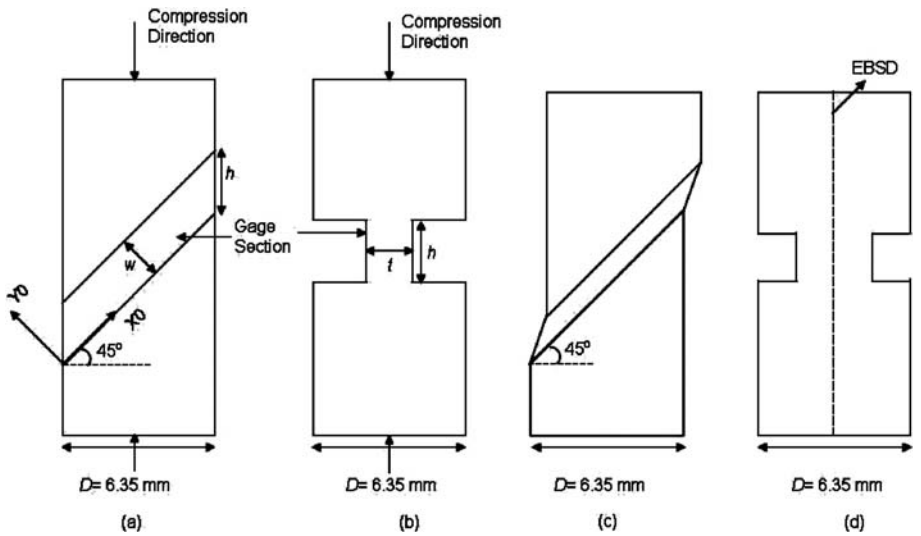


Fig. 1. Schematic of the SCS: (a) and (b) undeformed cross-sections with initial gage width $w = 1.06$ mm and initial gage thickness $t = 2.05$ mm, (c) and (d) cross-section after deformation. The initial gage height $h = \sqrt{2} \cdot w$. EBSD was performed on the inside surface after cutting the specimen by EDM.

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Applied

2. Experimental

2.1. Material, specimens and testing

A high-purity polycrystalline Ta rod of diameter (D) 6.35 mm was obtained from Accumet Materials Co., NY 10510, USA. The composition of the rod is given in Table 1. Such a high level of purity was achieved by our supplier through electron beam zone refining. Three 13 mm long cylindrical specimens were machined (from the rod) into SCS with the gage width (w) and the gage thickness (t) of 1.06 mm and 2.05 mm, respectively (Fig. 1). The samples were then deformed at different strain rates to the same strain.

Two of the Ta samples, referred from now on as Specimens 1 and 2, were compressed quasi-statically using a servohydraulic machine (MTS Model No. 11019) at strain rates of $10^{-3}\,s^{-1}$ and $10^{-1}\,s^{-1}$, respectively. The third sample, Specimen 3, was compressed dynamically using a split Hopkinson pressure bar [20], at a strain rate of $10^{+3}\,s^{-1}$. All of them were deformed to the same strain level of 0.25.

After the deformation, the samples were cut in half along their heights parallel to the gage section by electro-discharge machining (EDM), as shown in Fig. 1. The sample sections were then ground using SiC papers and electropolished using a solution of 10 ml HF and 90 ml H_2SO_4 in a bath with an applied voltage of 40 V for $\sim$1 min, in order to achieve microscopically-flat, mirror-like surface finish, for good electron back scattered diffraction (EBSD) characterization. The automated orientation maps (and corresponding textures) of the Ta samples were performed using an HKL electron backscattered diffraction (EBSD) system along with the Channel 5 software, attached to a high-resolution analytical scanning electron microscope (SEM, LEO 1550VP).

2.2. Data reduction

When the SCS is compressed, the gage section actually experiences a three-dimensional state of stress with the dominance of shear stress. Based on the past computational analysis done on the SCS by Rittel et al. [14, 15], the stresses and strains in the gage section are reduced to the equivalent von Mises stress (σ_e) and strain (ε_e). Quadratic constitutive relations for the Ta-SCS were developed, using the numerical procedure reported in [19], as follows:

$$\varepsilon_e = \varepsilon_Y + K_3\frac{(d-d_Y)}{h} + K_4\frac{(d-d_Y)^2}{h^2} \quad (1)$$

$$\sigma_e = K_1(1-K_2\varepsilon_e)\frac{P}{Dt} \quad (2)$$

where d is the vertical displacement of the gage section and P is the applied load. As shown in Fig. 1, the gage height h = 1.50 mm, the diameter D = 6.35 mm, the initial thickness t = 2.05 mm. The terms ε_Y and d_Y correspond to the strain and displacement values at the yield point, respectively. The parameters K_1, K_2, K_3 and K_4 were determined as 0.92, 0.15, 0.96 and −0.55 respectively, for the specific geometrical parameters of the SCS employed in this study [19].

Table 1. Chemical composition (weight ppm) of Ta found from the supplier, Accumet Materials Co., NY 10510, USA.

Element		Weight (ppm)	Element		Weight (ppm)
C	<	10	Ni	<	1
O	<	10	Si	<	10
N	<	10	Ti	<	1
H	<	5	W	<	6
Nb	<	19	Mo	<	2
Fe	<	3	Ta	>	999950

3. Result

3.1. Mechanical response

Curves a and b in Fig. 2 show the stress–strain response of Specimens 1 and 2 under quasi-static deformation. As the deformation rate is increased, there is a corresponding increase in the yield stress, with no significant increase in the strain hardening curve. This is typically observed for bcc metals when deformed from $10^{-6}\,s^{-1}$ to $1\,s^{-1}$ [21].

Curve c in Fig. 2 shows the stress–strain response of Specimen 3 deformed dynamically. As the strain rate is increased by nearly 6 orders of magnitude, the yield stress is more than doubled, from $\sim$250 MPa to $\sim$680 MPa. Additionally, the material shows no strain hardening up to a strain of 0.25, resulting in a nearly flat stress–strain curve. The initial oscillations of the curve at low strains are due to the non-equilibrium of the stress along the length of the SCS that is generally observed during dynamic deformation.

3.2. Microstructural characterization

3.2.1. Undeformed texture

As has been reported before for similar tests using the SCS, the deformation is restricted only in the gage section during the compression, and is dominated by shear [13, 14]. The rest of the specimen remains undeformed. This feature was

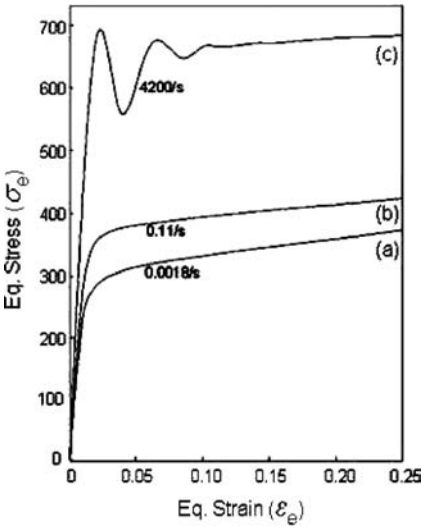


Fig. 2. Stress–strain curves of the three SCS Ta specimens deformed at various strain rates: (a) Specimen 1 at $1.8 \cdot 10^{-3}\,s^{-1}$, (b) Specimen 2 at $1.1 \cdot 10^{-1}\,s^{-1}$, and (c) Specimen 3 at $4.2 \cdot 10^{+3}\,s^{-1}$.

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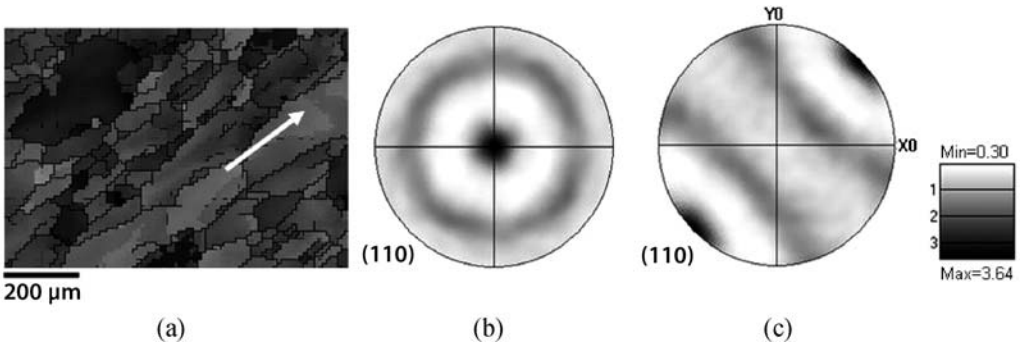


Fig. 3. (a) Euler-angle map of the undeformed section of SCS Ta with grains elongated along the rod length (indicated by the arrow). A step size of 10 µm was used and a grain boundary was designated for a neighboring orientation difference of more than 20°. (b) The corresponding (110) pole figure exhibiting a strong (110) fiber texture with the fiber axis along the specimen height, and (c) texture of the same region plotted in the $X0-Y0$ reference frame as evinced in Fig. 1.

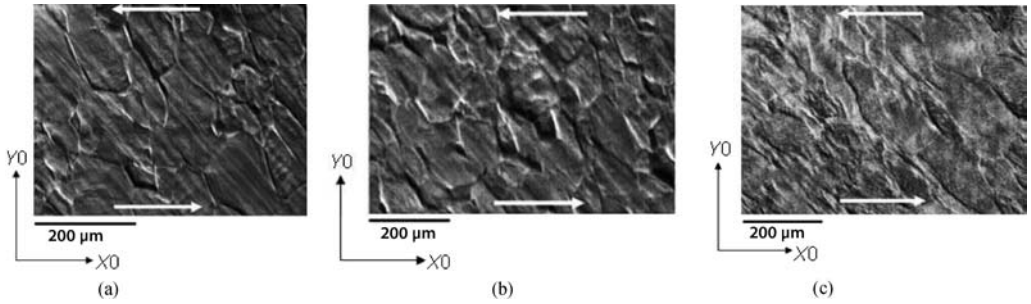


Fig. 4. SE-images of the mid-section of the sheared region of the SCS Ta with the arrows indicating the direction of shear: (a) Specimen 1 deformed at a strain rate of $1.8 \cdot 10^{-3}\,s^{-1}$, (b) Specimen 2 deformed at a strain rate of $0.11\,s^{-1}$, and (c) Specimen 3 deformed at a strain rate of $4200\,s^{-1}$.

verified (again) here by noticing that the crystallographic texture was the same in the undeformed cylindrical sections of all the three samples after compression, and was the same as that of the parent material. Figure 3a shows a representative undeformed microstructure, with the grains elongated along the length of the as-received Ta rod. Figure 3b shows the corresponding (110) pole figure of the undeformed section revealing a strong (110) fiber texture, plotted with the normal axis (i.e. axis through the center of the pole figure) along the compression direction. Figure 3c shows a variation in the depiction of the texture by choosing the axes along the gage section, $X0-Y0$ (Fig. 1).

3.2.2. Deformed microstructure

Figure 4a shows an SEM micrograph of the deformed grains in the gage region for Specimen 1, deformed quasi-statically at a strain rate of $1.8 \cdot 10^{-3}\,s^{-1}$. The grains appear sheared along the shear direction after the deformation. Also noticeable are the slip lines across the grains. Similarly, Fig. 4b shows the SEM micrograph of the deformed grains of Specimen 2, deformed quasi-statically at a strain rate of $0.11\,s^{-1}$, and finally, Fig. 4c shows the micrograph of the deformed grains of Specimen 3, deformed dynamically at a strain rate of $4200\,s^{-1}$. An HgCdTe IR detector (Judson Technologies, LLC) was used to measure the temperature rise in the gage section during the high-rate test and correspondingly, a maximum rise of 65 °C was found.

3.2.3. Deformation texture

Figure 5b and c shows the crystallographic texture of Specimen 1 in the gage section which was deformed to a strain of 0.25. The axis $X0-Y0$ was chosen in such a way that the $(+X0)$ axis is along the negative shear in the gage section. The Euler-angle map (Fig. 5a) of the deformed region producing the resulting texture is also shown adjacently. It can be seen that only a small fraction (i.e. 25 %) of the investigated area could be properly indexed due to the heavy deformation of the gage section[1]. In order to achieve a statistically more significant representation of the deformation texture in the gage region, Euler-angle data from many such individual EBSD scans near the mid-section of the deformed region were combined together and plotted as pole figures. Additionally, sample symmetry for 180° rotation about an axis normal to the $X0-Y0$ plane was invoked (i.e. keeping the coordinate system same). The resulting final texture in the form of contour plots is shown in Fig. 6. Such a procedure was similarly followed for Specimens 2 and 3. We notice that there is a significant change in texture of the gage section from its original undeformed texture (Fig. 3), after the region underwent a shear dominant deformation.

Figure 7 shows the texture of the deformed section of Specimen 2, quasi-statically deformed to the same strain as that of Specimen 1. And finally, the deformation texture of Specimen 3 is shown in Fig. 8 which was dynamically deformed to the same strain as Specimens 1 and 2. All of them had the same texture to begin with.

3.2.4. Texture comparison

By comparing the contour plots of Figs. 6 and 7, it is noticed that the high intensity regions are nearly at the same location (in the pole figure) for both (110) and (111) pole figures. Moreover, the values of maximum pole density at these locations are comparable to each other. Therefore,

[1] The percentage indexing could have been improved by scanning smaller areas with smaller step size.

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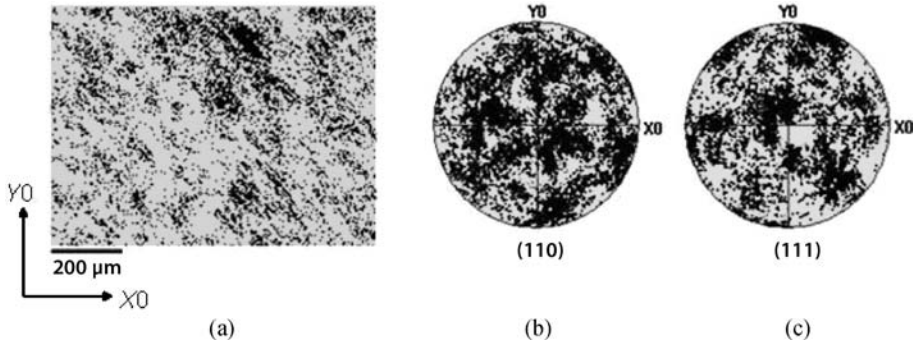


Fig. 5. (a) Euler-angle map of the mid-section of the gage region for Specimen 1, with dark colored pixels indicating a proper Kikuchi pattern indexing. An area of 1.0 mm by 0.6 mm was scanned with 5 µm step size resulting in 25 % pixel indexing. The texture of the region is represented by (110) and (111) pole figures in (b) and (c) respectively, using equal area projection.

we can claim that the deformation texture, qualitatively and quantitatively, remains unaffected with changes in strain rate during quasi-static loading.

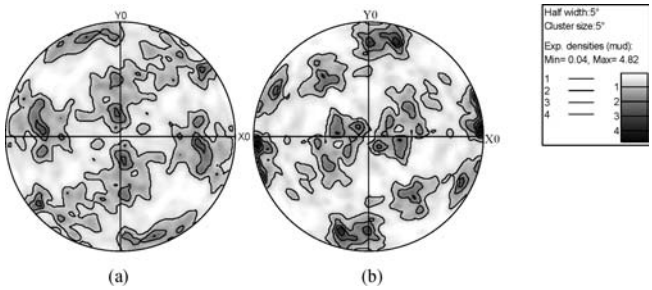


Fig. 6. (a) (110) pole figure and (b) (111) pole figure of the deformed grains in the mid-section of the sheared region of SCS Ta Specimen 1 in the $X0-Y0$ reference frame. Such a texture was generated by scanning a total area of 2.42 mm^2, with step size of 5 µm and around 25 % pixel indexing.

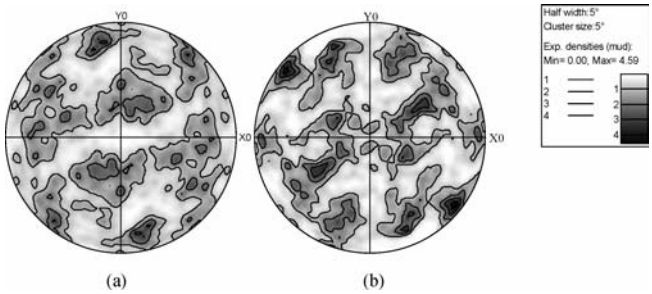


Fig. 7. (a) (110) pole figure and (b) (111) pole figure of the deformed grains in the mid-section of the sheared region of SCS Ta Specimen 2 in the $X0-Y0$ reference frame. Such a texture was generated by scanning a total area of 1.54 mm^2, with step size of 5 µm and around 25 % pixel indexing.

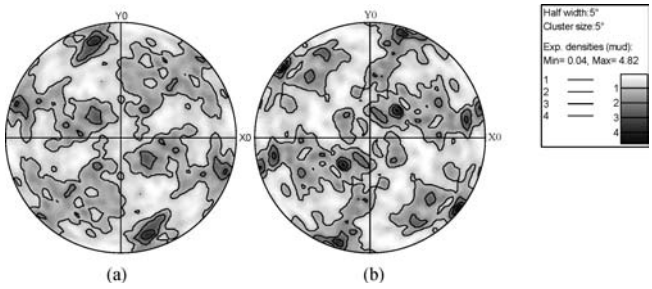


Fig. 8. (a) (110) pole figure and (b) (111) pole figure of the deformed grains in the mid-section of the sheared region of SCS Ta Specimen 3 in the $X0-Y0$ reference frame. Such a texture was generated by scanning a total area of 1.52 mm^2, with step size of 5 µm and around 25 % pixel indexing.

In the case of Specimen 3, the contour plots reveal an increase in the density of the clustering of the poles at specific locations, compared to its counterparts in Figs. 6 and 7. Qualitatively, however, its texture appears similar to that of Specimens 1 and 2.

These sets of observations allow us to believe that the deformation texture in bcc Ta is independent of the strain rate but with enhanced textural sharpness at high rates. According to the authors, no such experimental study has been done thus far. The only report of such non-dependence has been mentioned in the paper by Lee et al. [9] but without any experimental undertaking like ours.

Finally, it is also expected that such observation of strain rate independence on deformation texture for Ta is independent of the mode of deformation.

4. Discussion

In this section we try to explore the reasons behind the strain rate independence of deformation texture for bcc Ta. In order to understand the textural similarity at these widely different strain rates, the corresponding mechanical response was looked upon for analysis (Fig. 2). From the stress–strain curves at $1.8 \cdot 10^{-3}$ s^{-1} and 0.11 s^{-1}, we notice that the strain hardening remains unaffected with the increase in the strain rate. The strain hardening exponent (n), defined by $\mathrm{d}(\ln \sigma_e)\,/\mathrm{d}(\ln \varepsilon_e)$, is around 0.13 for Specimen 1 and 0.11 for Specimen 2, in the linear strain regime. At high rates, due to the adiabatic nature of the deformation, the material experiences thermal softening leading to a decrease in the flow stress. The exponent n is around 0.06 in the linear range, an order lower than its value in the quasi-static regime. However, such a large decrease in the strain hardening exponent is found to have no significant qualitative effect on its deformation texture as it appears nearly the same as that at low strain rates.

In order to understand the extent to which the deformation texture is being controlled by strain hardening, we base our analysis on the dislocation theory. It is known that strain hardening is primarily governed by the interaction between mobile dislocations and its surrounding dislocation environment i. e. higher strain hardening indicates that larger stresses are needed to glide the mobile dislocations across, against the local resistive force of its dislocation surroundings. As the strain rate of the deformation process increases, the stress needed for the initiation of dislocation motion increases (resulting in higher yield strength). Comparatively, bcc materials are highly strain rate sensitive, with the strain rate sensitivity coefficient (m) being ~ 0.05 in the present case.

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Similarity in the hardening behavior in Specimens 1 and 2 (n being nearly the same) seems to indicate similarity in the arrangement of dislocations in the material, when deformed to the same strain level. An extension or an assumption can be made here from the similarity in macroscopic dislocation arrangement to the arrangement at a crystal level that the dislocation structure also remains the same within individual crystals (having same orientation) of Specimens 1 and 2.

The total deformation (or strain) experienced by a crystal is composed of crystal rotation and plastic shear. The amount of shear (produced due to slip activity on various slip systems) is related to mobile dislocation density and its velocity (through obstacles) by Orowan's equation. Therefore, it may be considered that the amount of shear for two identically oriented crystals would be the same if they have the same dislocation arrangement, experiencing the same overall strain. In continuation, similarly oriented crystals of Specimens 1 and 2 would undergo the same rotation, under same local strain and shear[2]. Collectively, this leads us to the observation of texture similarity between Specimens 1 and 2.

In the case of Specimen 3, which was dynamically deformed, the temperature rise within the sample led to strain softening by favoring dislocation rearrangement/annihilation, and thereby allowing free movement of mobile dislocations. However, the temperature increase in this case was only $\sim 65\,^{\circ}\mathrm{C}$ which might not have contributed much towards a complete change in the orientation of a crystal, but more towards creating sharper grains/sub-grain boundaries. Consequently, a gradual variation in local orientation within a grain or grain boundary may be modified into a more distinct orientation change. This steers towards formation of a sharper texture. Looking closely, we do observe high intensity orientation spots in the (111) pole figure of Specimen 3 (Fig. 8) compared to a more diffused orientation distribution for Specimens 1 and 2 (Figs. 6 and 7). Qualitatively however, the texture at high strain rates appears very similar to the corresponding textures at low strain rates. Such similar behavior of texture sharpening was also noticed in α-Fe when deformed at a strain rate of $\sim 2500\ \mathrm{s}^{-1}$ [22].

5. Conclusions

The effect of strain-rate, up to ($10^{+3}\ \mathrm{s}^{-1}$), on deformation texture of Ta has been investigated at room temperature by means of EBSD. The following conclusions can be drawn for bcc Ta from this study:

- There is no qualitative change in the deformation texture when the strain rate is increased from quasi-static to dynamic regime.
- Temperature rise during dynamic deformation is found to sharpen its deformed texture.
- A direct correspondence is found between strain hardening rate and deformation texture. Similarity in hardening seems to dictate similarity in the texture at different strain rates, keeping all the other parameters the same.

[2] As their crystal orientations are the same, the same slip systems are being activated.

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Correspondence address

Dr. Abhishek Bhattacharyya
Department of Aeronautics and Mechanical Engineering, California Institute of Technology
Pasadena, CA 91125, USA
Tel.: +1 626 395 4748
Fax: +1 626 449 2677
E-mail: ab03@caltech.edu

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